

ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ENGINEERING MATHEMATICS I

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 1st year Engineering Mathematics I paper 1 · practice a.

Written by Academic Central for practice, following first-year Engineering Mathematics syllabuses at South African universities (BEng Mechatronics, Electrical and Mechanical Engineering). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method that differs from the one shown, and answers that differ only through rounding.

QUESTION 1

1.1 $\log_2[x(x-2)] = 3$ ✓ (4)
$x^2 - 2x = 8$, so $x^2 - 2x - 8 = 0$ ✓
$(x-4)(x+2) = 0$ ✓
$x = 4$ ($x = -2$ is rejected because $\log_2$ of a negative number is undefined) ✓

1.2 $3x + 5 = A(x+3) + B(x+1)$ ✓ (4)
$x = -1$: $2 = 2A$, so $A = 1$ ✓
$x = -3$: $-4 = -2B$, so $B = 2$ ✓
$(3x+5)/((x+1)(x+3)) = 1/(x+1) + 2/(x+3)$ ✓

1.3.1 Domain: $x \geq 2$ ✓ Range: $y \geq 0$ ✓ (2)

1.3.2 $y = \sqrt{(x-2)} \rightarrow x = y^2 + 2$, so $f^{-1}(x) = x^2 + 2$ ✓ domain $x \geq 0$ ✓ (2)

1.4 $R = \sqrt{(3^2 + 4^2)} = 5$ ✓ (3)
$\tan\alpha = 4/3$, $\alpha = 53.13°$ ✓
$3\sin\theta + 4\cos\theta = 5\sin(\theta + 53.13°)$; maximum value = 5 ✓

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QUESTION 2

2.1.1 $4 + 2j$ ✓ (1)

2.1.2 $(3+4j)(1-2j) = 3 - 6j + 4j - 8j^2$ ✓ $= 11 - 2j$ ✓ (2)

2.1.3 $(3+4j)(1+2j)/((1-2j)(1+2j))$ ✓ $= (3 + 6j + 4j - 8)/5$ ✓ $= -1 + 2j$ ✓ (3)

2.2 $r = \sqrt{(9+16)} = 5$ ✓ $\theta = \tan^{-1}(4/3) = 53.13°$, so $z_1 = 5\angle 53.13°$ ✓ (2)

2.3 $1 + j = \sqrt{2}\angle 45°$ ✓ (3)
$(1+j)^8 = (\sqrt{2})^8 \angle(8 \times 45°) = 16\angle 360°$ ✓ $= 16$ ✓

2.4 $z = (-2 \pm \sqrt{(4 - 20)})/2 = (-2 \pm 4j)/2$ ✓ $z = -1 + 2j$ or $z = -1 - 2j$ ✓ (2)

2.5 $2e^{\wedge}(j\pi/2)$ ✓ $= 0 + 2j = 2j$ ✓ (2)

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QUESTION 3

3.1.1 $(x-2)(x+2)/(x-2) = x + 2$ ✓ $\rightarrow 4$ ✓ (2)

3.1.2 Divide by x^2: $(3 + 1/x^2)/(2 - 1/x)$ ✓ $\rightarrow 3/2$ ✓ (2)

3.1.3 0/0, so differentiate top and bottom: $\lim (e^x - 1)/(2x)$, still 0/0 ✓ (2)
Again: $\lim e^x/2 = 1/2$ ✓

3.2 $f'(x) = \lim (h \to 0)$ of $[f(x + h) - f(x)] / h$ ✓ (4)
$= \lim [(x + h)^2 - x^2] / h = \lim (2xh + h^2) / h$ ✓
$= \lim (2x + h)$ ✓
$= 2x$ ✓

3.3.1 Product rule ✓: $dy/dx = 3x^2 e^{2x} + 2x^3 e^{2x}$ ✓ $= x^2 e^{2x} (3 + 2x)$ ✓ (3)

3.3.2 $dy/dx = 2x / (x^2 + 1)$ ✓✓ (2)

3.3.3 Quotient rule ✓: $dy/dx = (x \cos x - \sin x) / x^2$ ✓✓ (3)

3.4 $2x + 2y (dy/dx) = 0$ ✓ (3)
$dy/dx = -x/y$ ✓
At (3; 4): $dy/dx = -3/4$ ✓

3.5 $V = x(12 - 2x)^2$ ✓ (4)
$dV/dx = (12 - 2x)(12 - 6x) = 0$, so $x = 2$ ($x = 6$ gives no box) ✓
$V = 2 \times 8^2 = 128$ cm³ ✓
$d^2V/dx^2 = 24x - 96 = -48 < 0$ at $x = 2$, so it is a maximum ✓

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QUESTION 4

4.1 $x^3 - 2x^2 + 5x + C$ ✓✓ (one mark for the powers, one for the constant C) (2)

4.2 Let $u = x^2 + 1$, $du = 2x\,dx$ ✓ (3)
$\int u^4\,du = u^5/5 + C$ ✓ $= (x^2 + 1)^5 / 5 + C$ ✓

4.3 $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$ ✓ (4)
$\int x e^x\,dx = x e^x - \int e^x\,dx$ ✓✓
$= x e^x - e^x + C$ ✓

4.4 $\lim (b \to \infty)$ of $[-1/x]$ from 1 to b $= \lim (-1/b + 1)$ ✓ $= 1$ ✓ (2)

4.5 $x^2 = 2x$ gives $x = 0$ and $x = 2$ ✓ (5)
Area $= \int$ from 0 to 2 of $(2x - x^2)\,dx$ ✓
$= [x^2 - x^3/3]$ from 0 to 2 ✓
$= 4 - 8/3$ ✓ $= 4/3$ square units ✓

4.6 $\int [1/(x + 1) + 2/(x + 3)]\,dx$ ✓ $= \ln|x + 1| + 2 \ln|x + 3| + C$ ✓✓ (3)

4.7 $V = \pi \int$ from 0 to 4 of $(\sqrt{x})^2\,dx$ ✓ $= \pi [x^2/2]$ from 0 to 4 ✓ $= 8\pi$ cubic units ✓ (3)

4.8 $e^x \approx 1 + x + x^2/2! + x^3/3!$ ✓✓ (3)
e^0.1 ≈ 1 + 0.1 + 0.005 + 0.000167 = 1.1052 ✓

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QUESTION 5

5.1 AB = [2 0 ; 1 5] ✓✓ (one mark for each correct row) (2)

5.2 det A = 2 × 3 − 1 × 1 = 5 ✓ (3)
$A^{-1} = (1/5)$ [3 −1 ; −1 2] ✓✓

5.3 $[x ; y] = A^{-1}$ [5 ; 10] ✓ (3)
$x = (1/5)(15 - 10) = 1$ ✓ $y = (1/5)(-5 + 20) = 3$ ✓

5.4 $1[(-1)(1) - (4)(0)] - 2[(3)(1) - (4)(2)] + 0$ ✓ (3)
$= -1 - 2(-5)$ ✓
$= 9$ ✓

5.5.1 $a \cdot b = (2)(1) + (-1)(2) + (2)(-2)$ ✓ $= -4$ ✓ (2)

5.5.2 $|a| = 3$, $|b| = 3$, $\cos\theta = -4/9$ ✓ $\theta = 116.39°$ ✓ (2)

5.5.3 $a \times b = i[(-1)(-2) - (2)(2)] - j[(2)(-2) - (2)(1)] + k[(2)(2) - (-1)(1)]$ ✓ (3)
$= -2i + 6j + 5k$ ✓✓

5.5.4 $-2(x - 1) + 6(y - 0) + 5(z - 2) = 0$ ✓ (2)
$-2x + 6y + 5z = 8$ ✓

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TOTAL: 100

END OF MARKING GUIDELINE